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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (CBCS) Degree Examinations
December 2025

IS1102-2- INTRODUCTION TO DATA SCIENCE
(For ISE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. _____ is the process of rendering to date aspects of the world that have never been quantified before.
A) Datification
B) Data Mining
C) Data warehouse
D) Data Wrapping
2. _____ can be gathered by passive or by active methods
A) Outcome
B) Decision
C) Scalability
D) Data
3. _____ analytics allows decisions to be taken in response to future events, not only reactively
A) Suggestive
B) Predictive
C) Informative
D) Decisive
4. The values of any DataFrame can be retrieved as a Python array by calling its _____ attribute.
A) data
B) values
C) size
D) index
5. _____ are data samples with a value that is far from the central tendency
A) wrapping
B) oddity
C) mining
D) outliers
6. A _____ interval is a range of values that best represents some parameter of interest
A) time
B) confidence
C) flexible
D) correlated
7. The development of traditional statistics was limited by the _____ of computational resources
A) scarcity
B) increase
C) varying
D) similarity
8. The p-value is the _____ of observing data at least as favourable to the alternative hypothesis as the current dataset, if the null hypothesis is true.
A) flexibility
B) associativity
C) probability
D) cumulative
9. The edge between two _____ can be directed or undirected
A) nodes
B) routers
C) graphs
D) network
10. The _____ of a node is the number of edges that connect to it.
A) link
B) degree
C) weight
D) size



11. Which data mining task is used to predict categorical outcomes?
 - A) Clustering
 - B) Classification
 - C) Regression
 - D) Association
12. In which phase of data mining do you clean and preprocess data?
 - A) Data Understanding
 - B) Data Preparation
 - C) Model Building
 - D) Model Evaluation
13. Which of the following is a benefit of data mining in business?
 - A) Better Decision Making
 - B) Cost Reduction
 - C) Improved Customer Satisfaction
 - D) Customer Retention
14. Which of the following is a major issue in data mining?
 - A) Data Input
 - B) Data Encoding
 - C) Data Visualization
 - D) Scalability
15. What is "data reduction" in data mining?
 - A) The process of removing redundant data
 - B) The process of cleaning and transforming data
 - C) The process of summarizing data
 - D) The process of storing data in a compressed format
16. What is the main challenge in applying data mining techniques to real-world data?
 - A) Data quality and variability
 - B) Insufficient computing resources
 - C) Lack of powerful algorithms
 - D) Too much data
17. Which of the following is a supervised learning algorithm?
 - A) Apriori method
 - B) K means method
 - C) KNN method
 - D) Decision Tree method
18. Which supervised learning algorithm is known for its ability to handle both classification and regression tasks?
 - A) Random Forest
 - B) K-nearest neighbors (KNN)
 - C) Support Vector MachineS(SVM)
 - D) Linear Regression
19. Which algorithm is suitable for handling multi-class classification problems in supervised learning?
 - A) Multinomial Logistic Regression
 - B) Naive Bayes
 - C) Support Vector MachineS(SVM)
 - D) Decision Tree
20. Which supervised learning algorithm is used for time series forecasting?
 - A) Decision Tree
 - B) Multinomial Logistic Regression
 - C) Recurrent Neural Networks (RNN)
 - D) Linear Regression

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the various stages in a data science project, providing examples for each stage.	8	L*2	1	1
	b) Define Ego-Networks and explain their significance in network analysis. Discuss the methodologies and goals of Community Detection within a complex system.	8	L1	2	1
2.	a) Analyze the significance of exploratory data analysis and describe methods for data summarization and outlier treatment.	8	L3	2	1
	b) Discuss the applications of data science in at least three fields, showing how it transforms each industry.	8	L2	1	1
3.	a) Discuss estimation techniques and compare the sample mean and variance with their estimated values, including their relevance in statistical analysis.	8	L2	1	2

- b) What is Social Network Analysis? Describe the concept of Centrality in graphs and explain how PageRank works as a measure of centrality.

8 L1 2 2

Unit – II

4. a) Define Data Quality in the context of data preprocessing. Describe the four major tasks in data Pre-processing.
 b) Describe the steps involved in Data Warehouse Design and Usage. How is a Data Warehouse implemented, and what are the key considerations for effective implementation?
5. a) Define data mining and discuss its major challenges, benefits, and real-world applications.
 b) Explain the concepts and implementation of data warehousing, including basic concepts, modelling, and OLAP systems.
6. a) Explain the various data mining tasks and the phases involved in a typical data mining project.
 b) Differentiate between OLAP (OnLine Analytical Processing) and OLTP (OnLine Transactional Processing).

8 L1 3 3

8 L2 4 3

8 L1 3 3

8 L2 4 3

8 L2 3 4

8 L2 4 4

Unit – III

7. a) Explain how Support Vector Machines (SVM) operate and discuss their suitability for different data science problems.
 b) Compare training, validation, and test sets in supervised learning, explaining their roles in model development.
8. a) Discuss clustering, focusing on similarity measures, distance metrics, and methods for evaluating clustering quality.
 b) Introduce unsupervised learning and differentiate it from supervised learning, providing examples where each is used.

8 L2 5 4

8 L2 5 4

8 L2 5 5

8 L2 5 5

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
